What's AI actually costing the planet?

What California is, and isn't doing about the environmental side of the tech boom.

Every AI query has an energy bill

Every time you use ChatGPT or any AI-powered app, a data center is running somewhere in the background. These are massive facilities packed with servers that never turn off. They need a lot of electricity to run, and even more to stay cool. Right now, data centers use about 1.5% of all electricity worldwide.

2x by 2030

Data center electricity use is expected to double by 2030, growing more than 4x faster than any other industry.

Keeping these centers from overheating takes water too; around 17 billion gallons were used in U.S. data centers alone in 2023. That figure could grow nearly five times over the next decade. In California, where droughts are a regular problem, that's more than an abstract concern.

It's more than just electricity consumption

Data centers also affect nearby communities. Backup generators emit air pollutants—linked to cardiovascular and respiratory disease—into nearby neighborhoods. Clusters of data centers can also cause temperature increases of 1.5-2.4°C within a year, up to 4.5 kilometers away. Cities like San Francisco, where AI-related business activities are growing rapidly, are relevant.

There's been real legislative activity

California has been pretty active on AI regulation. Results are mixed. As of July 2026, here's where the key bills stand:

SB 1047 — VETOED 2024
Would have required AI companies to test their most powerful models and disclose detailed safety information before release.

SB 53 — SIGNED INTO LAW 2025
A lighter version. Companies still have to publish safety policies and report on their AI systems, just with fewer restrictions.

SB 886 — PENDING
Would require large data centers to prepay for 15 years of clean energy infrastructure.

SB 1168 — PENDING
Would fine data centers for excessive energy consumption.

However, these laws all have gaps

Each of these bills is a meaningful step. But they all have real limitations.

Transparency bills don't address environmental impact. SB 53 is about AI safety. It doesn't require companies to report how much electricity or water they use. And because so much of AI's energy comes through complicated supply chains, tracking the true environmental footprint is genuinely hard.

The energy bills have loopholes. SB 886's threshold only kicks in above a certain size, meaning companies could split into smaller nearby facilities to dodge it. But clustering smaller data centers together might actually make local heat worse. And SB 1168's fines could just become a routine business expense for large tech companies.

Both bills only apply in California. Nothing stops a company from moving operations to Nevada or Texas, where regulations are weaker.

A few directions worth considering

Standardized reporting
If all AI companies had to disclose energy and water use in a consistent format, cities could actually track the impact. Right now, San Francisco's plan to reach net-zero by 2040 doesn't mention AI at all. That's a real gap.

Cleaner backup power
Replacing diesel generators with alternatives like battery storage or hydrogen fuel cells would reduce air pollution in communities near data centers.

Updated climate planning
Cities need plans that actually account for AI's growth. Tricky, since AI's trajectory is hard to predict, but pretending it's not a factor isn't really an option anymore.

None of these are simple fixes, as stricter rules could push companies to other states and cleaner energy costs more upfront. Real tradeoffs exist between innovation and protecting communities.

Why should students care?

Most of us only see the screen. Not the data centers, electricity, water, heat, or backup generators behind it. That does not mean students should stop using AI. It means we should understand the tradeoffs. The choices being made now about AI infrastructure will affect climate goals, local communities, electricity demand, water use, and future policy.

Students are not just AI users. We are future voters, workers, researchers, and decision-makers. If AI is going to shape our future, we should have a voice in how we can benefit from AI without ignoring its environmental cost.

This explainer is adapted from my research paper, *AI Energy and Environmental Policy in San Francisco*, which includes citations and analysis. Key sources include the International Energy Agency, Pew Research Center, Next 10, Marinoni et al., San Francisco climate planning, and legislation SB 1047, SB 53, SB 886, and SB 1168. AI is shaping our future. What tradeoffs should we care about? Share your perspective in this form.